

Aaron Modiyil Joseph

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PROFILE

MSc Data Science and BSc Mathematics graduate with a solid foundation in statistics, machine learning, financial analysis, and risk assessment. Experienced in data analysis, financial reconciliations, reporting, and process improvement in a fast-paced operational environment. Proficient in Python, R, SQL, Excel, Power BI, and statistical modelling methodologies, with a track record of transforming complex datasets into actionable insights. An IFoA Student Member with a great passion for actuarial science, risk management, financial markets, and data-driven decision-making.

EDUCATION

MSc. Data Science with Advanced Research (Distinction), University of Hertfordshire, Hatfield, UK (Jan 2023 – Mar 2025)

- Developed analytical, modelling, and data visualization skills through large-scale data projects and applied research.
- Thesis: Predicting Movie Ratings and Analyzing Feature Importance Using Tags and Genres
- Relevant Modules: Statistical Modelling, Python, SQL, Data Analysis, Data Visualization, Time Series Analysis, Machine Learning, AI Ethics.

BSc. Mathematics (Distinction), St. Berchmans College, Kerala, India (Jun 2019 – Sept 2022)

- Strong foundation in Statistics, Probability, Hypothesis Testing, Linear Algebra, Calculus, Differential Equations, and Economics.
- Developed rigorous quantitative reasoning and problem-solving skills applicable across analytics, finance, actuarial science, and risk management.

PROFESSIONAL MEMBERSHIPS & CERTIFICATIONS

- Institute and Faculty of Actuaries (IFoA) – Student Member
- CS1(Actuarial Statistics) sat in April 2026 – Results pending
- Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure AI Fundamentals

TECHNICAL SKILLS

- Programming:** Python (Pandas, NumPy, Scikit-Learn), R, SQL, Excel (Pivot Tables, VBA, Power Query), Power BI, Tableau.
- Data Analysis & Reporting:** Excel, Power BI, Tableau, Matplotlib, Seaborn.
- Quantitative & Statistical Skills:** Statistical Modelling, Regression Analysis, Time Series Analysis, Predictive Analytics, Risk Analysis, Data Validation, Financial Reconciliation, Trend Analysis, Reporting, Process Automation.
- Other Tools:** GitHub, Jupyter Notebook, Microsoft Azure, LaTeX

PROFESSIONAL EXPERIENCE

Night Auditor and Guest Service Assistant | Holiday Inn Express (IHG Brand), Watford, UK (June 2023 – Present)

- Reconcile daily financial transactions and operational records, ensuring reporting accuracy and data integrity.
- Investigate and resolve discrepancies across systems while maintaining strong financial controls.
- Produce audit reports and operational summaries within strict reporting deadlines.
- Analyze occupancy, revenue, and customer trends through Power BI dashboards to support business decision-making.
- Automated welcome letter generation for IHG members using Excel and Word, incorporating tier-level benefits and alphabetical ordering, replacing a manual system.
- Collaborated with finance, operations, reservations, and customers to resolve issues and maintain service quality.
- Handle confidential information while maintaining high standards of professionalism and accuracy.
- Work effectively in a fast-paced environment where accuracy and timely completion of tasks are critical.

Mathematics Tutor | Universal Tuition Centre, Kerala, India

(May 2021 - Jan 2023)

- Delivered mathematics and statistics tuition to students with diverse learning needs.
- Simplified complex quantitative concepts through structured problem-solving approaches.

- Developed strong communication, presentation, and mentoring skills.

PROJECTS

Financial Market Forecasting

- Developed forecasting models using historical market data to analyze share price movements.
- Conducted time-series analysis and data preprocessing to identify market trends and predictive factors.
- Assessed model performance using statistical error metrics against baseline forecasting methods and sensitivity analysis to evaluate forecasting reliability.
- Built Power BI dashboards to communicate investment insights and model results.

Comparative Study of ARIMA and LSTM Models on Time Series Data

- Compared ARIMA and LSTM forecasting models on seasonal and volatile financial datasets.
- Investigated seasonality, volatility, and temporal trends in time-series data.
- Compared forecasting accuracy using statistical error metrics.
- Evaluated model suitability across different volatility environments, highlighting the importance of selecting appropriate modelling approaches for risk forecasting.

Comparative Study of Clustering Models

- Compared GMM and K-Means clustering models on a large rental dataset.
- Investigated relationships between multiple quantitative variables through exploratory analysis.
- Evaluated clustering performance using statistical and visual assessment techniques.
- Demonstrated how model selection influences segmentation outcomes and risk interpretation.

Predicting Movie Ratings and Analyzing Feature Importance Using Tags and Genres

- Developed predictive models in Python to analyze relationships between user-generated tags, genres, and movie ratings within a large-scale dataset.
- Performed exploratory data analysis and feature engineering on large datasets.
- Applied statistical analysis to identify rating patterns and user behaviour trends
- Evaluated model performance using statistical error metrics and visual analysis.

VOLUNTEERING & ACTIVITIES

- Active committee member of the Physics, Astronomy and Mathematics Society, Sim Racing Society, and Anime Society at the University of Hertfordshire.
- Organized events and collaborated with diverse teams to deliver successful student activities.
- Led the Rubik's Cube competition during college events, demonstrating leadership, organization, and teamwork skills.
- Participated in and won mathematics competitions throughout school and university.

ADDITIONAL INFORMATION

Languages - Fluent in *English*, Hindi, Malayalam.

Hobbies and Interests - Personal Finance | Financial Markets | Rubik's Cubes | Playing Keyboard | Football | Cricket | Motorsports | Sim Racing | Movies | Music